Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 October 2026
Team ID	NA
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Tableau calculation language (SUM, AVG, COUNT, Top-N)	Aggregates sales, range, CO_2 and ratings without coding.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV file (ev_market_test.csv) + Tableau extract (.hyper)	Simple file source; extract is fast and required by Tableau Public.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau Data Connection – Text file	Loads all 100,000 rows directly into Tableau.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop (data source checks, sorting, filters)	Data type, null and duplicate checks; Top-10 filters; sorting.

5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop / Tableau Public	8 worksheets, dashboard and story.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public (web), GitHub	Public link for users; GitHub stores files and documentation.